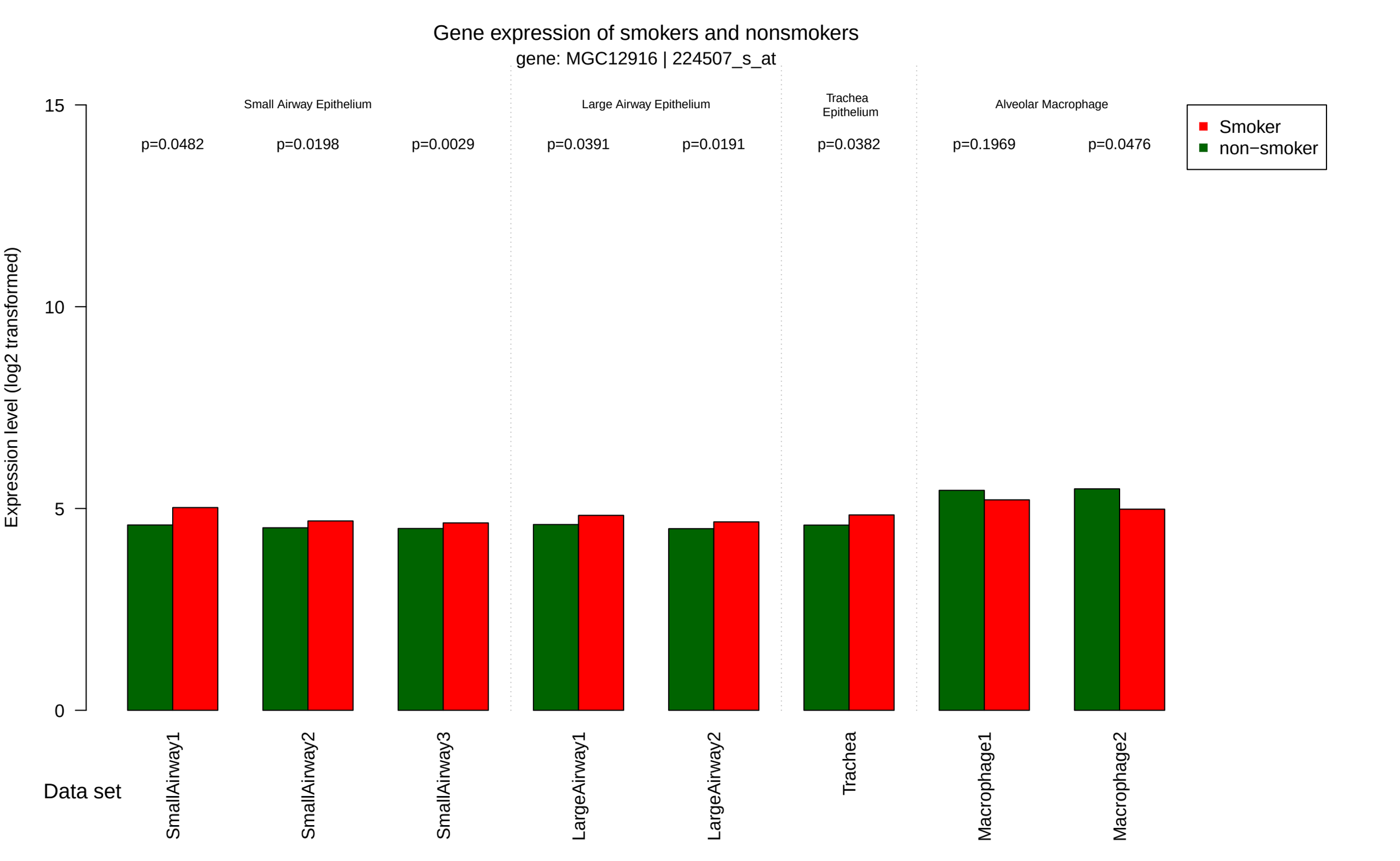
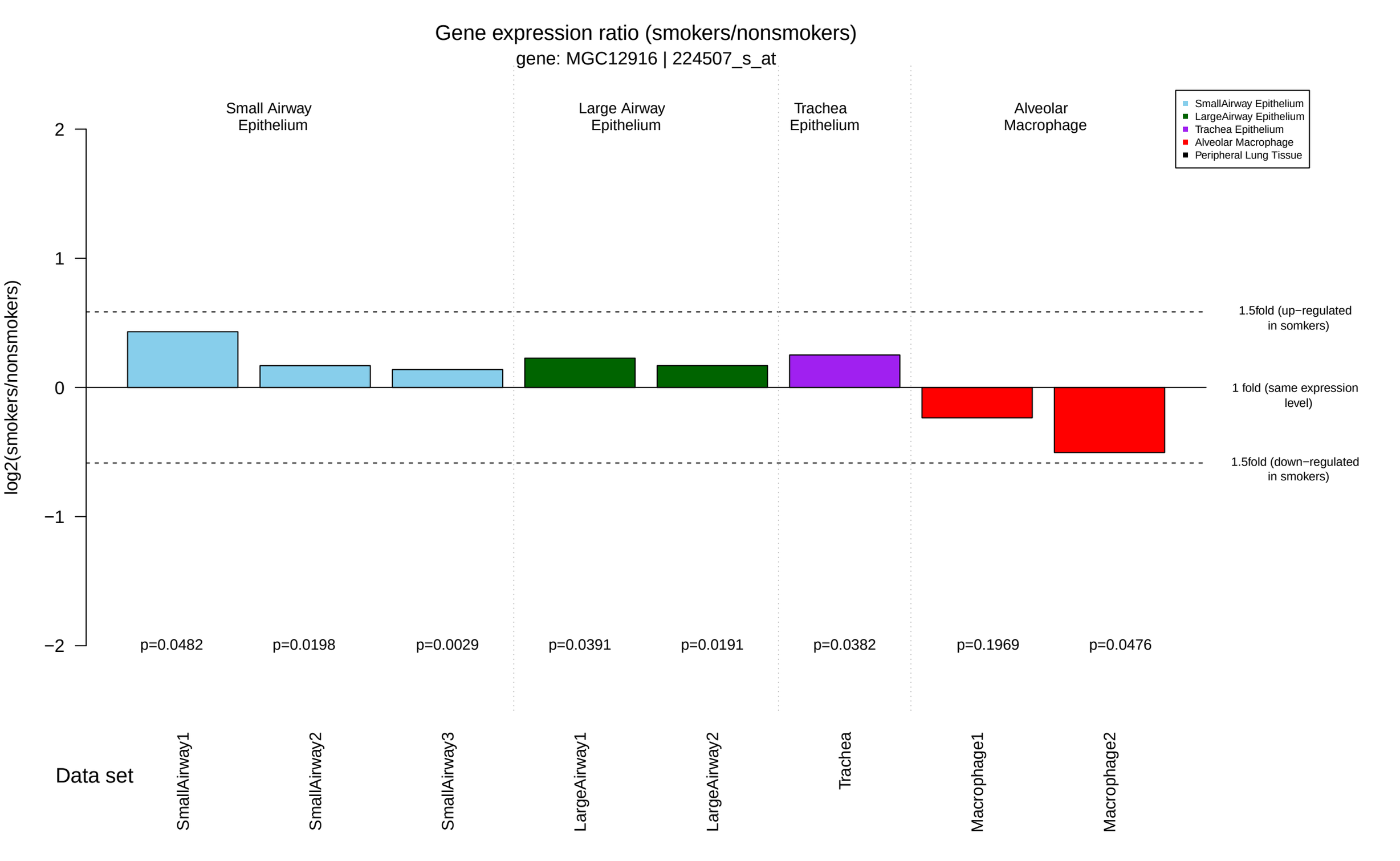
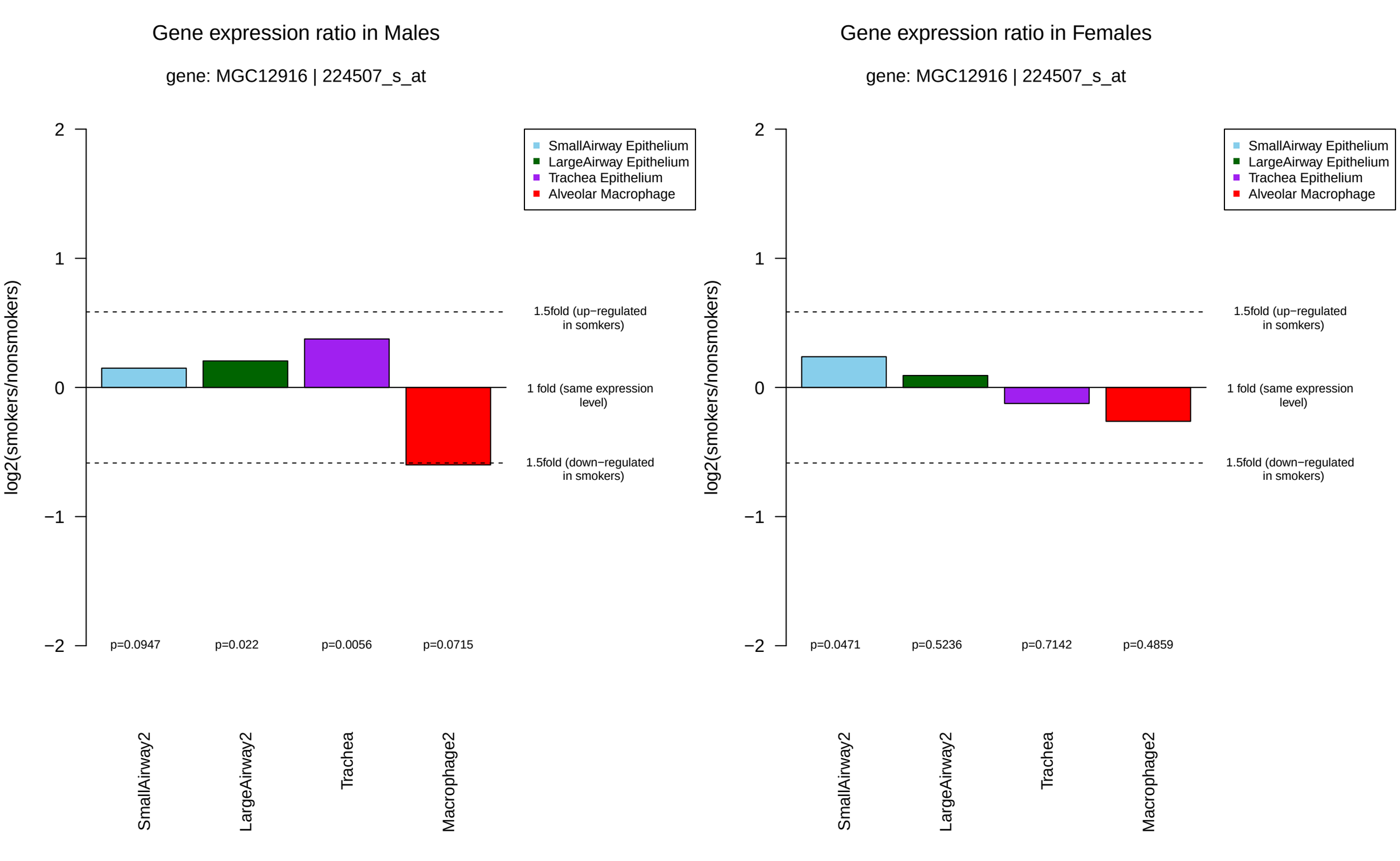
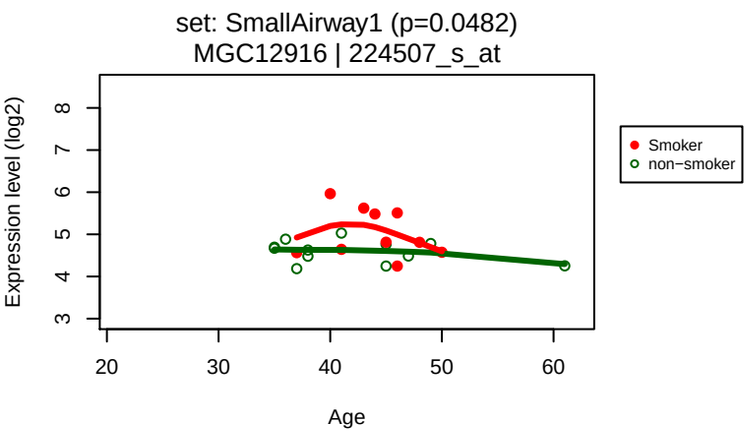
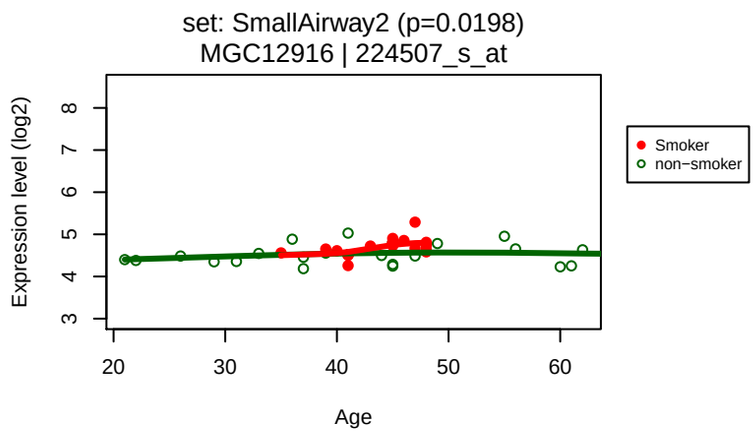
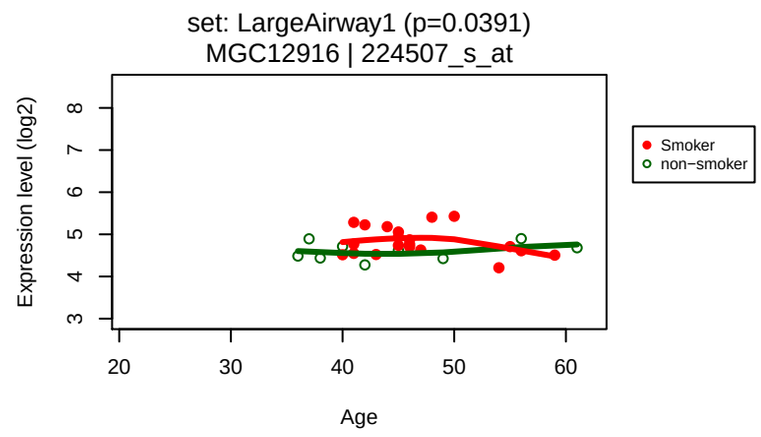

Gene expression of smokers and nonsmokers
gene: MGC12916 | 224507_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0482
p=0.0198
p=0.0029
p=0.0391
p=0.0191
p=0.0382
p=0.1969
p=0.0476
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

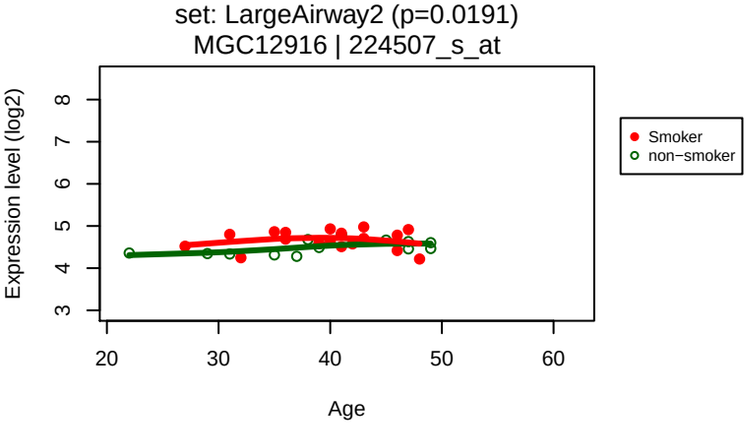

Gene expression ratio (smokers/nonsmokers)
gene: MGC12916 | 224507_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0482
p=0.0198
p=0.0029
p=0.0391
p=0.0191
p=0.0382
p=0.1969
p=0.0476
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

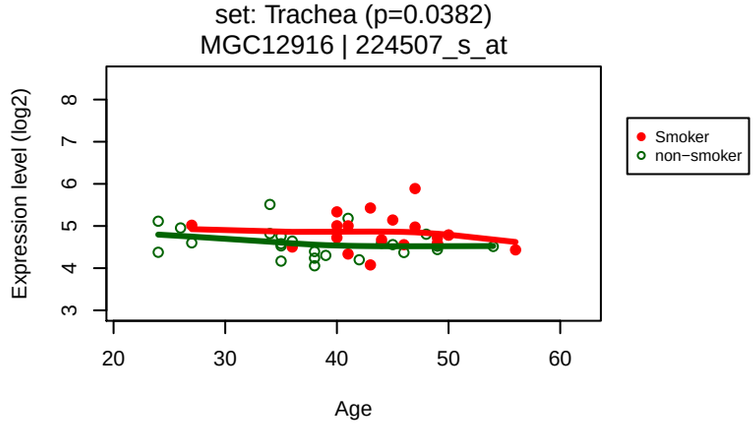

Gene expression ratio in Males
gene: MGC12916 | 224507_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0947
p=0.022
p=0.0056
p=0.0715
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MGC12916 | 224507_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0471
p=0.5236
p=0.7142
p=0.4859
SmallAirway2
LargeAirway2
Trachea
Macrophage2

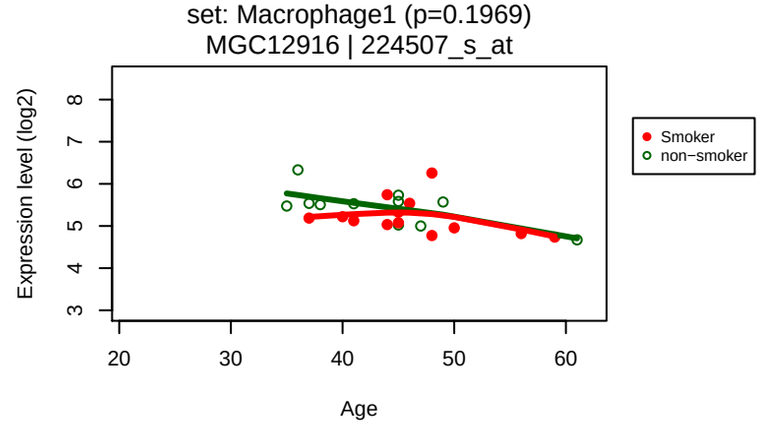

set: SmallAirway1 (p=0.0482)
MGC12916 | 224507_s_at
Expression level (log2)
Age
Smoker
non−smoker

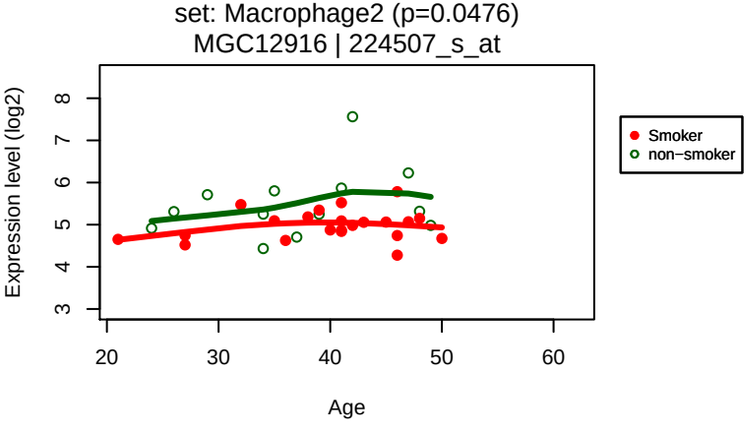

set: SmallAirway2 (p=0.0198)
MGC12916 | 224507_s_at
Expression level (log2)
Age
Smoker
non−smoker

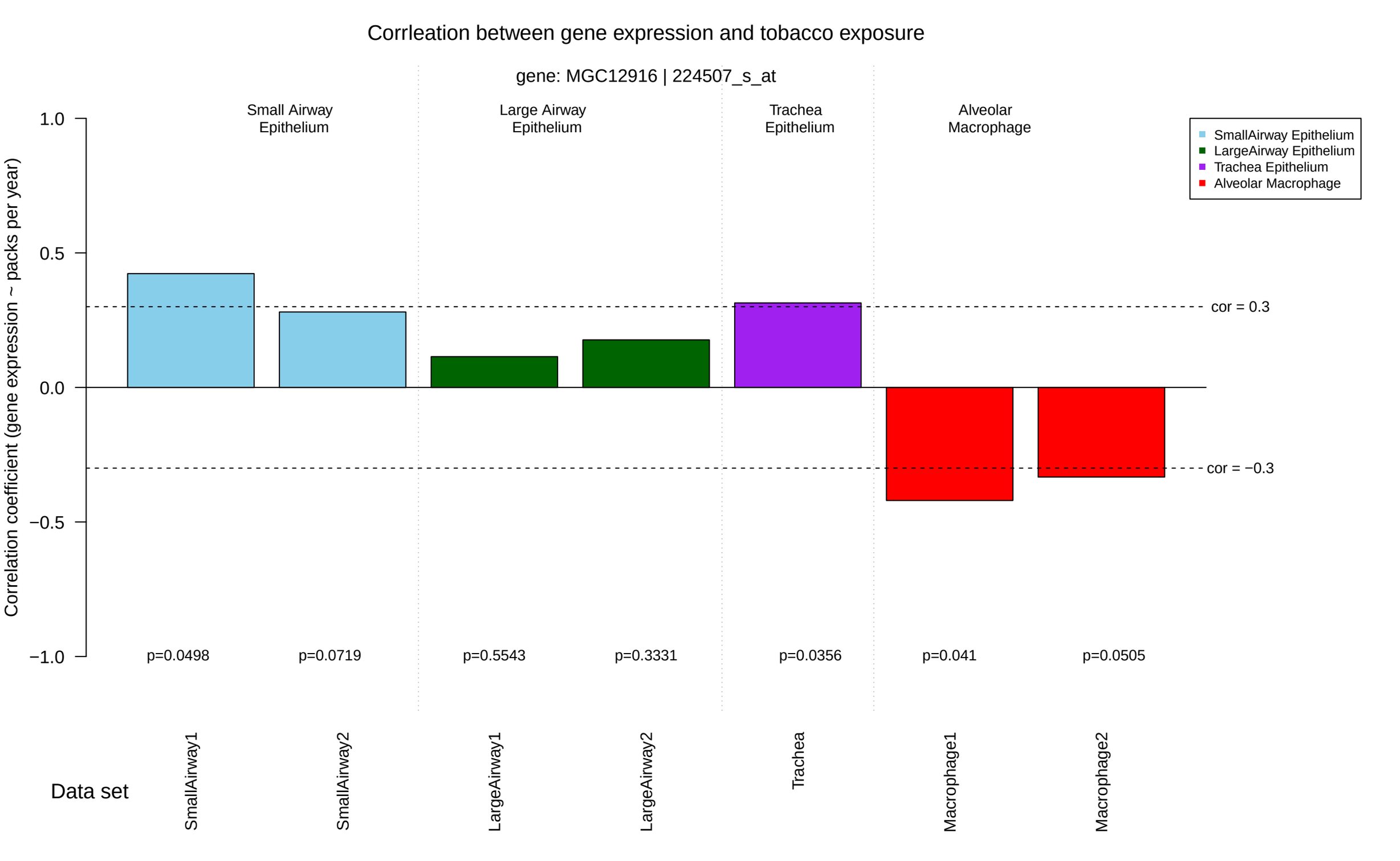

set: LargeAirway1 (p=0.0391)
MGC12916 | 224507_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0191)
MGC12916 | 224507_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0382)
MGC12916 | 224507_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1969)
MGC12916 | 224507_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0476)
MGC12916 | 224507_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MGC12916 | 224507_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0498
p=0.0719
p=0.5543
p=0.3331
p=0.0356
p=0.041
p=0.0505
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2